

THÉO HUET

Theoretical Physics Master's student focused on modeling noise and interactions in quantum circuits, seeking to contribute to the EDA pipeline at Alice & Bob.

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theo-huet TheoHUETQC



EXPERIENCES

Master's Research Internship

Collège de France

03/2026 - 07/2026 Paris (75)

- Compare the performance of the **tensor network (MPO)** and **Pauli chain propagation** methods for the time evolution of a local operator in the presence of **external noise** (implemented in Julia). Under the supervision of M. De Nardis,
- Related to the GitHub repository: Quantum-ManyBody-Sim-Methods

Participant in the Quant25 Master's summer school

Max-Planck Institute

09/2025 Dresden, Germany

- "From Quantum Matter to Quantum Computers",
- A week of conferences on **quantum computing (QEC, Tensor Network, Reinforcement learning)** and condensed matter physics.

Bachelor's Research Internship

Laboratory of Theoretical Physics and Modeling (LPTM)
CY Cergy Paris University

05/2024 Cergy (95)

- Analysis of repeated measurement on quantum systems, re-demonstrating the **Quantum Zeno Effect** using Cook's Proposition under the supervision of M. Fratino.

AWARDS

Bourse d'excellence Master 2

Institute of Science and Technology, CY Cergy Paris Université

EDUCATION

Master's in Theoretical Physics

CY Cergy Paris Université

2024 - 2026 Cergy (95)

Specialization: Quantum Error Correction, Numerical Simulation.

Double Bachelor's Degree in Mathematics and Physics

CY Cergy Paris Université

2021 - 2024 Cergy (95)

General Baccalaureate Maths-Physique,
Advanced Maths option

Agricultural and Horticultural High School

2021 Saint-Germain-en-Laye (78)

PROJECTS AVAILABLE ON MY GITHUB

FEM-based electromagnetic analysis (Supervised Study)

06/2026 - 08/2026 Personal

Monte-Carlo Physics Simulations

02/2026 - 03/2026 Master 2

- Ising Model, Diffusion QMC and Stochastic Series Expansion (SSE) in Python.

Quantum Error Mitigation with ZNE

02/2026 Personal

- Zero-Noise Extrapolation example using **Qiskit**.

Machine-Learning predicting sleep quality with physiological data

01/2026 - 02/2026 Master 2

- Using Linear Regression, Random Forest and Neural Networks with TensorFlow/Keras.

SKILLS

Python (Qiskit, NumPy, SciPy, Matplotlib)

Julia (ITensors, ITensorMPS, PauliPropagation)

Git, GitHub

LaTeX, Jupyter Notebook

LANGUAGES

French Native

English B1-B2

INTERESTS

Diorama (in Lego and 3D Printing to create realistic scenes)

Video games (Developing my own Game/Physics engines)

Home improvement (plumbing, electricity, renovation)

Climbing (weekly)